

Cybersecurity



Course-End Project





Enhancing Windows Security with Logon Monitoring and Reporting

Objective

To enhance Windows security by implementing advanced audit configurations, including failed login analysis, command logging, and real-time reporting, leveraging PowerShell automation and web-based tools for effective monitoring and mitigation of potential breaches



Problem Statement and Motivation



Real-time scenario:

TechSecure, a mid-sized enterprise, faces increasing cyber threats targeting its Windows-based systems, particularly through unauthorized login attempts and malicious command executions. To address this, TechSecure implements an advanced security configuration strategy.

They use Event Viewer to analyze failed logins, leveraging Event ID 4625 to identify potential intrusions and track suspicious IP addresses. By deploying group policies, they enable command logging and process auditing, ensuring every executed command is documented. Automation through PowerShell scripts allows real-time tracking of successful logins and automated HTML reporting for a failed attempt.

This proactive approach not only mitigates security risks but also provides comprehensive insights into user activities. By strengthening its Windows security posture, TechSecure ensures the safety of its systems and enhances its credibility as a secure and reliable enterprise.

Industry Relevance

The following tools used in this project serve specific purposes within the industry:

1. **Event Viewer:** Provides detailed system and user activity logs, enabling administrators to monitor login attempts, identify suspicious behavior, and strengthen system security
2. **PowerShell:** Automates log analysis, report generation, and event tracking, offering real-time monitoring and efficient management of security configurations
3. **Group Policy Management:** Enables advanced audit configurations and detailed logging of processes, enhancing visibility into user actions and improving threat detection
4. **HTML Reporting:** Generates automated, easy-to-read reports for failed logon attempts, aiding in quick analysis and response to security events



Tasks

The following tasks outline the process of enhancing Windows system security through logon monitoring, event auditing, and automated reporting:

1. Set up audit policy to do logon failure analysis
2. Command login configuration
3. Track successful logon
4. Create a web-based reporting tool



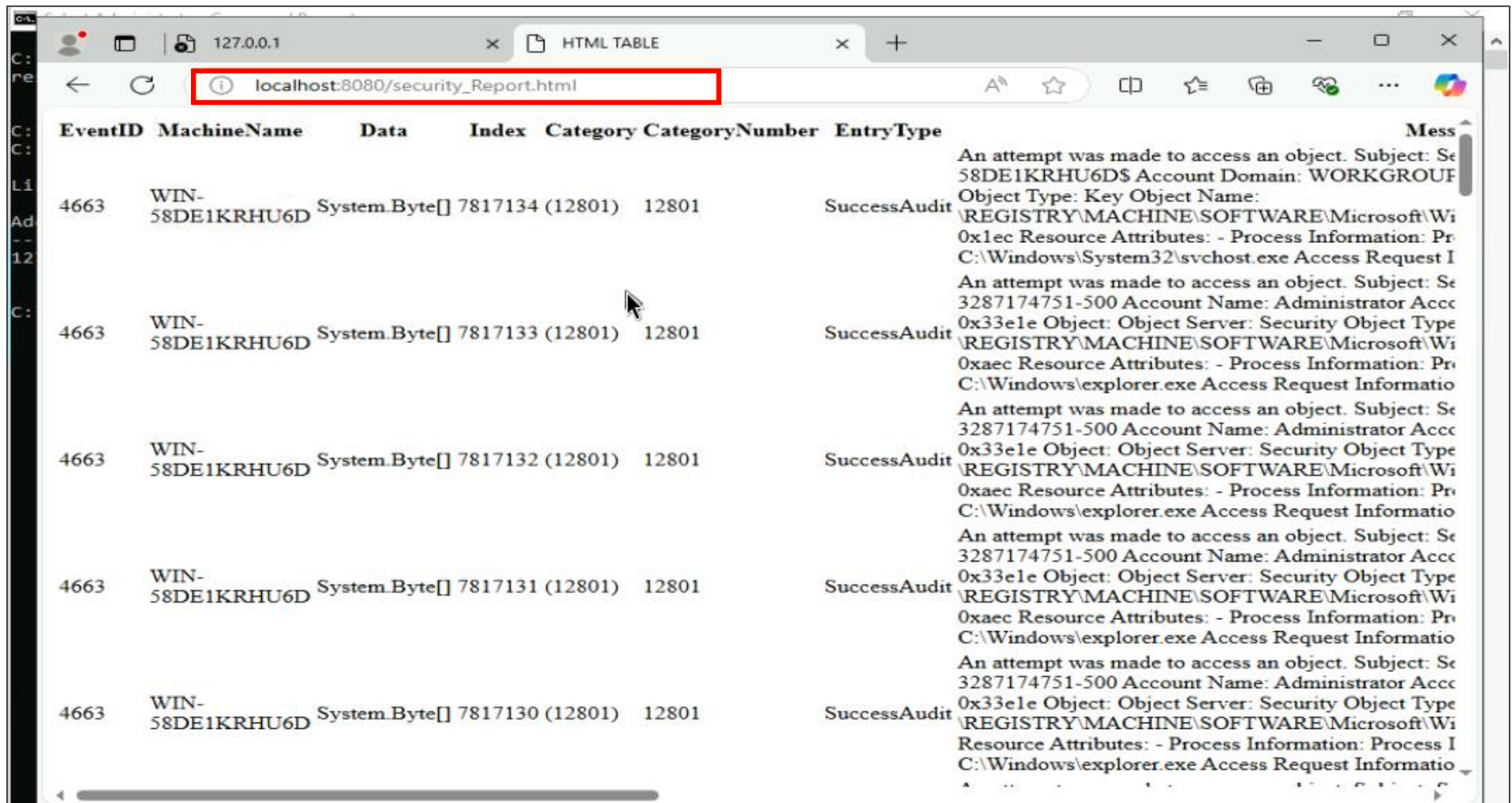
Project References

- **Task 1:** Lesson 07
- **Task 2:** Lesson 02
- **Task 3:** Lesson 07
- **Task 4:** Lesson 05



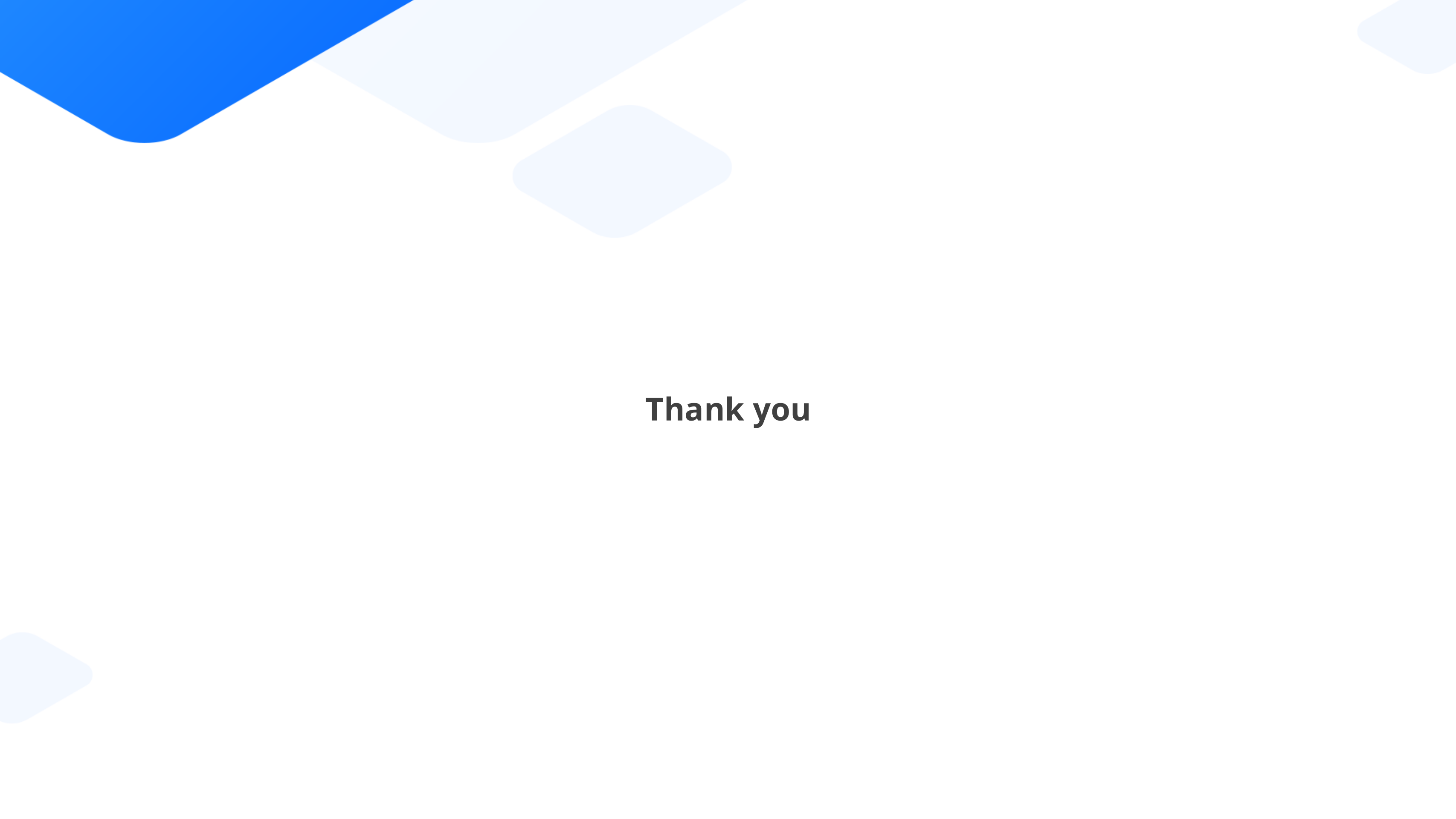
Output Screenshots

The security configurations are successfully implemented, monitoring logon attempts and generating real-time reports for both successful and failed logins as shown below:



The screenshot shows a web browser window with the address bar displaying 'localhost:8080/security_Report.html'. The browser has two tabs: '127.0.0.1' and 'HTML TABLE'. The main content area displays a table with security event data. The table has columns: EventID, MachineName, Data, Index, Category, CategoryNumber, EntryType, and Message. The data shows five successful audit events (EventID 4663) for the machine 'WIN-58DE1KRHU6D'. Each event is a 'SuccessAudit' entry with a 'Category' of 'System.Byte[]' and a 'CategoryNumber' of '12801'. The messages describe attempts to access registry objects.

EventID	MachineName	Data	Index	Category	CategoryNumber	EntryType	Message
4663	WIN-58DE1KRHU6D	System.Byte[]	7817134 (12801)	12801		SuccessAudit	An attempt was made to access an object. Subject: Se 58DE1KRHU6D\$ Account Domain: WORKGROU... Object Type: Key Object Name: \REGISTRY\MACHINE\SOFTWARE\Microsoft\Wi 0x1ec Resource Attributes: - Process Information: Pr C:\Windows\System32\svchost.exe Access Request I
4663	WIN-58DE1KRHU6D	System.Byte[]	7817133 (12801)	12801		SuccessAudit	An attempt was made to access an object. Subject: Se 3287174751-500 Account Name: Administrator Acco 0x33e1e Object: Object Server: Security Object Type \REGISTRY\MACHINE\SOFTWARE\Microsoft\Wi 0xaec Resource Attributes: - Process Information: Pr C:\Windows\explorer.exe Access Request Informatio
4663	WIN-58DE1KRHU6D	System.Byte[]	7817132 (12801)	12801		SuccessAudit	An attempt was made to access an object. Subject: Se 3287174751-500 Account Name: Administrator Acco 0x33e1e Object: Object Server: Security Object Type \REGISTRY\MACHINE\SOFTWARE\Microsoft\Wi 0xaec Resource Attributes: - Process Information: Pr C:\Windows\explorer.exe Access Request Informatio
4663	WIN-58DE1KRHU6D	System.Byte[]	7817131 (12801)	12801		SuccessAudit	An attempt was made to access an object. Subject: Se 3287174751-500 Account Name: Administrator Acco 0x33e1e Object: Object Server: Security Object Type \REGISTRY\MACHINE\SOFTWARE\Microsoft\Wi 0xaec Resource Attributes: - Process Information: Pr C:\Windows\explorer.exe Access Request Informatio
4663	WIN-58DE1KRHU6D	System.Byte[]	7817130 (12801)	12801		SuccessAudit	An attempt was made to access an object. Subject: Se 3287174751-500 Account Name: Administrator Acco 0x33e1e Object: Object Server: Security Object Type \REGISTRY\MACHINE\SOFTWARE\Microsoft\Wi Resource Attributes: - Process Information: Process I C:\Windows\explorer.exe Access Request Informatio



Thank you